The Assessment Question Paper and Assessment Answer Booklet must not be removed from the assessment room. Both must be handed in at the end of the assessment.

SIM Global Education

WRITTEN TEST
CONTINOUS ASSESSMENT (MAIN)

(DIPLOMA IN INFORMATION TECHNOLOGY)

Date	**:**	**23 October 2025**
Module	**:**	**ITSD002 Problem Solving**
Time	**:**	**10:00 – 11:30**
Duration	**:**	**1 hour 30 minutes**
Total no. of pages (including this cover page)	**:**	**6**

INSTRUCTIONS TO CANIDATES:

1. This assessment carries a total of 100 marks. Answer **ALL** questions:

   Section A (50 marks): 2 Structural Questions
   Section B (50 marks): 2 Application Questions

2. Answer **ALL** questions in the answer booklets provided. Fill in the cover page carefully. **You will receive a zero (0) score if your name is filled incorrectly**.

3. All answers must be clearly written. Begin each question on a fresh page.

4. The use of non-programmable calculators is **NOT** permitted in this assessment.

5. This is a closed-book assessment. No unauthorised materials are allowed in the assessment venue.

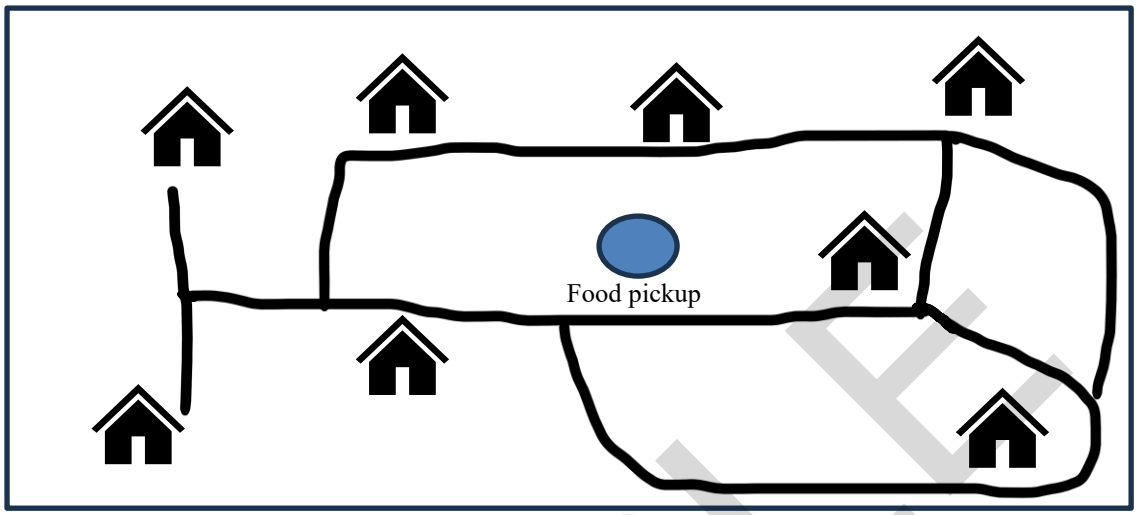
At the end of the assessment:
Please ensure that you have written your name on each answer booklet used: failure to do so will mean that your work cannot be identified. If you have used more than one answer book, please tie them together with the string provided.

SIM GLOBAL EDUCATION RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

SAMPLE

SECTION A

Question 1



A food delivery cyclist has to deliver the same food items to the same 8 houses every day. However, he can only carry 3 houses' worth of items each time, then has to go back to the pickup point again. He wants to find the best route to take so that he can save time and effort.

Required:

(a) Describe the Brute Force problem solving method and how it can be used by the cyclist to find the best route.

(10 marks)

(b) Describe the Heuristics problem solving method and how it can help the cyclist to find the best route.

(10 marks)

(c) The first step to the Development Cycle is *Analyse the problem.* What are the 5 other steps?

(5 marks)

SAMPLE

Question 2

Given a list of data: 19, 22, 34, 38, 45, 50 stored in an array (or list), answer (a) and (b). State clearly the values of the indices/positions and the values of the array during the process of sorting with relevant pictures or diagrams.
(Note: you do not need to write the algorithm/code)

Required:

(a) Describe Linear Search for the value of '38' from start to finish.
(10 marks)

(b) Describe Binary Search for the value '22' from start to finish.
(10 marks)

(c) State **TWO (2)** differences between Linear search and Binary search.
(5 marks)

SAMPLE

SECTION B

Question 3

NoodleHouse is creating a simple app to allow users to order food. The users have an option to select delivery or self-pickup. There are 3 sizes of noodles – Small ($2.50), Medium ($3.50), Large ($5). Delivery costs an extra $5. There is no extra cost for self-pickup.

Required:

(a) Create a Flowchart that allows the user to first choose the noodle size, the quantity (e.g. 3 packets), and delivery or self-pick-up. Output the total cost.

(10 marks)

(b) Write in Pseudocode a program that allows the user to choose the same options as in (a). In addition, the program should give a 10% discount on orders where total amount is more than $20. After ordering, the user should have an option "Do you want to repeat a new order (yes or no)". If user enter "yes", the program should repeat.

(15 marks)

SAMPLE

Question 4

For this question, the Food Ordering section of the program will be developed. The prices and noodle types are fixed. Delivery Fee is $5, but free if total is more than $20. Also assume that the user will input a correct number for the item.

Below is a sample run of the program.

[*user inputs are in* ***BOLD***]

```
---- CC Noodle Menu ----
1) Fishball Noodles ($4.80, $5.80, $6.80)
2) Beef Noodle Soup ($6, $7, $8)
3) Beef Noodle Dry ($6.50, $7.50, $8.50)

Enter number of item to order: 2
Do you want Small (S), Medium (M), or Large (L)?: S
Do you want to add another item (y/n)? y

Enter number of item to order: 3
Do you want Small (S), Medium (M), or Large (L)?: L
Do you want to continue (y/n)? y

Enter number of item to order: 1
Do you want Small (S), Medium (M), or Large (L)?: M

Do you want to continue (y/n)?: n


Order total: $20.3
Delivery fee: $0

Do you want to add new order (y/n)?: y

---- CC Noodle Menu ----
1) Fishball Noodles ($4.80, $5.80, $6.80)
2) Beef Noodle Soup ($6, $7, $8)
3) Beef Noodle Dry ($6.50, $7.50, $8.50)

Enter number of item to order: 2
Do you want Small (S), Medium (M), or Large (L)?: L
Do you want to add another item (y/n)? n

Order total: $8
Delivery fee: $5

Do you want to add new order(y/n): n
```

SAMPLE

ORDER SUMMARY
Total number of orders: 2
Total amount earned (incl delivery): $33.3

Required:

Code the program in Python. Do indicate useful comments in your codes, and take note that indentation is crucial.

(25 marks)

---END OF PAPER---

SAMPLE